

Backbone Cabling and Topology

Backbone Distances

Intra-building backbone runs between telecommunications rooms and equipment rooms. For twisted-pair copper, the backbone is limited to 90 metres between cross-connects. For single-mode fibre, the intra-building backbone may reach 2,000 metres. Inter-building backbones can exceed 2,000 metres when using single-mode fibre.

- Copper backbone: 90 m
- Single-mode intra-building: 2,000 m
- Single-mode inter-building: >2,000 m

Star Topology

The hierarchical star is the mandatory topology for commercial building cabling. Each horizontal cabling segment home-runs to a telecommunications room. No intermediate splices or bridges are permitted without a cross-connect.